Exploring Vision Transformers for Polarimetric SAR Image Classification

Hongwei Dong, Lamei Zhang, *Senior Member, IEEE*, and Bin Zou, *Senior Member, IEEE*

***Abstract*— As one of the most popular topics in polarimetric synthetic aperture radar (PolSAR) community, PolSAR image classification has always been an important way for PolSAR applications. Constructing representations is the most critical part of PolSAR image classification. With the maturity of deep learning technique, many data-driven PolSAR representation methods have been proposed, most of which are based on convolutional neural networks (CNNs). Despite some achievements, the bottleneck of CNN-based methods may be related to the locality induced by their inductive biases. Considering this problem, the state-of-the-art method in natural language processing, i.e., transformer, is introduced into PolSAR image classification for the first time. Specifically, a vision transformer (ViT)-based representation learning framework is proposed in this article, which covers both supervised learning and unsupervised learning. For supervised learning, we use self-attention to replace convolution, which shifts the focus from the information in local neighborhoods to the long-range interactions between each pixel. Beyond supervised learning, we introduce an improved contrastive-based strategy to implement simple unsupervised representation learning. Compared with CNN and its variants, ViT constructs more global representations by explicitly modeling the relationship between each pixel, so as to improve the classification performance. Experimental results on four widely used PolSAR image datasets indicate that the representation obtained by the ViT-based methods is better for PolSAR image classification, whether supervised (up to about 5% accuracy improvement) or unsupervised (up to about 4%). In addition, we also prove the robustness of ViT to the initial input form. These discoveries may arouse rethinking of the dominance of CNNs in PolSAR image classification.**

***Index Terms*— Deep learning, polarimetric synthetic aperture radar (PolSAR) image classification, unsupervised learning, vision transformer (ViT).**

I. Introduction

POLARIMETRIC synthetic aperture radar (PolSAR) has become one of the most advanced sensors in the field of remote sensing because of its all-time and all-weather Earth observation capability. PolSAR uses different polarization channels to obtain complex-valued images to distinguish the structure, direction, and composition of the target. Such rich information is of great value to civil fields, such as land development [1], agriculture [2], and environmental change monitoring [3]. PolSAR image classification, as an important way to utilize PolSAR data, has received widespread attention.

As the central part of PolSAR image classification, the construction of PolSAR representations has been the focus of attention in recent years. Before feature extraction, speckle removal [4]–[6] is essential due to the speckle noise caused by the unique imaging mechanism of PolSAR system. Classic PolSAR feature extraction methods are mainly based on scattering mechanisms and statistical analysis. Among them, the most widely used method is target decomposition [7], whose advantage is that most of the results have clear physical explanations. Based on the basic scattering models, Freeman and Durden [8] calculated the contributions of surface, double bounce, and volume scatter to the total power. Subsequently, many improved methods with four scattering mechanisms, e.g., helix [9], refined volume [10], asymmetric [11], and multiple scattering models [12], [13], were proposed. Since it is difficult to evaluate polarimetric features separately [14], they are often combined with PolSAR image classification problems. Therefore, some reliable land cover classification techniques have been proposed [15]–[17], which realized the effective utilization of polarimetric features. However, due to the complexity of PolSAR echo imaging mechanism, the hand-crafted polarimetric features are still not satisfactory enough, and exploring better PolSAR representations and classifiers is still an open problem.

In recent years, deep learning [18] has been extensively studied in remote sensing images [19]. Different from classical methods, the concept of deep learning is to automate the feature engineering process. This data-driven feature extraction framework is also called representation learning, which is expected to be a substitute for handcrafted feature extractors. For the representation learning in image domain, convolutional neural networks (CNNs) are usually used as the carrier [20]. Zhou *et al.* [21] explored the application of CNNs in the PolSAR image classification for the first time. The 6-D polarimetric features were represented by two convolution layers and classified by two dense layers. The competitive experimental results provide confidence for the follow-up research.

Various supervised CNN-based PolSAR classification studies are springing up. Since which form of PolSAR data is the best input for CNNs is still unknown, Chen and Tao [22] studied the input with roll-invariant and rotation polarimetric features. Complete polarimetric information of scattering matrix was encoded as the input of a fully convolutional

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network [23]. Zhang *et al.* [24] extracted the amplitude and phase information of PolSAR data and constructed the matching network architectures. High-dimensional input data composed of multiple polarimetric features with feature selection networks have also been verified to be a feasible way [25]–[27]. In addition to using polarimetric features as the input, some studies are based on a polarimetric coherency matrix. Dong *et al.* [28] introduced the 3-D convolution to extract both spatial- and channel-wise information. In order to acquire the phase of the elements in coherency matrix, Zhang *et al.* [29] proposed to convert a real-valued model into a complex-valued CNN architecture. Tan *et al.* [30] proposed a complex-valued 3-D architecture to extract information from both spatial and scattering dimensions simultaneously. Model capacities also have a profound impact on the results of classification, and improvements can be achieved by optimizing topologies of basic CNNs, including but not limited to adding multipath [31], [32], residual connection [33], and manifold regularization [34]. Dong *et al.* [35] proposed a PolSAR-tailored neural architecture search method to improve the hyperparameters in a fixed network topology. Generative-based learning models have also been introduced to boost the performance [36]. Graph convolutional networks have been introduced and modified to extract multiscale PolSAR image features [37], [38] or implement semisupervised learning [39], so as to improve the classification performance. Generally speaking, the development of supervised PolSAR classification is in the ascendant, and the rising results [40] indicate that it has a certain potential.

The high cost of PolSAR image labeling results in the lack of supervision information, so the importance of unsupervised PolSAR representation learning is gradually discovered. Classical unsupervised methods mainly focus on clustering. Lee *et al.* [16] proposed a complex Wishart distribution-based maximum-likelihood classifier to implement unsupervised classification. Similarly, spectral clustering with a revised Wishart distance was applied to realize unsupervised PolSAR segmentation [41]. Autoencoders can achieve unsupervised representation learning based on reconstruction loss [42] or greedy layer-wise training [43], but the performance is ordinary and the computation is expensive. Recently, inspired by the pretraining mechanism in natural language processing [44], a new way of unsupervised image representation learning is proposed, namely, self-supervised learning (SSL) [45]. Different from learning with pseudo-labels [46], SSL follows the supervised learning paradigm, but the supervision is provided by the data itself. Contrastive learning is currently the most mainstream SSL method, which is motivated by measuring the similarity between two or more CNNs processed data augmentations to obtain high-level representations. Zhang *et al.* [47] first introduced the concept of contrastive learning into the PolSAR community and proposed a customized sample selection method to narrow the application gap. Similar to the idea of contrastive multiview coding [48], a mutual information-based SSL model was proposed in [49] to learn representation from unlabeled PolSAR data. Although it is not widespread now, similar research may become mainstream in the near future.

From the related work, it can be concluded that convolutional architectures are still dominant in PolSAR image representation learning and classification. It is well known that local connection and weight sharing are the key inductive biases of CNNs. However, the local nature of convolution is not always favorable. The convolution operation tries to get features from a local neighborhood, which blocks the acquisition of global information. Therefore, convolution is suitable for processing the grid data whose spatially adjacent elements are relevant and spatially distant ones are irrelevant. Although the locality of CNN features will decrease as the receptive field increases, high correlation between the pixels contained in the input pixel-centric PolSAR image patches [21] may cause the features that contain a lot of local information to be suboptimal. Recently, the state-of-the-art language model, i.e., transformer, was proven to be able to capture the global correlation in image domain [50]. In vision transformer (ViT), tokens (words) in the language are replaced by image patches, and self-attention mechanism [51] is used to model the long-range interactions [52]. Intuitively, this paradigm may be a good alternative to the convolutional architecture in the PolSAR image classification.

Based on the above analysis and inspired by previous works, a ViT-based representation learning framework is proposed in this article to effectively capture global information from pixel-centric PolSAR image patches, thereby improving the classification performance. Compared with the current popular CNN-based methods, our advantages are summarized as follows. First, unsupervised deep learning is included in the proposed framework, which is not available in conventional methods. Second, the features with more global information obtained by ViT-based methods are suitable for the task of patch-based PolSAR representation learning and classification. Third, the proposed framework is simple and effective, which is reflected in that on the one hand, the ViT backbone can get satisfactory supervised performance with different initial forms and, on the other hand, the unsupervised version can obtain high-quality representations without the involvement of complex components. The main contributions are given in the following.

1) The capacity of ViT for PolSAR image classification is explored for the first time. Pixels of PolSAR image patches are regarded as tokens of the language, and the long-range interactions of them can be established through the self-attention mechanism in the model. A learnable class token and multilayer perceptron (MLP) are used to integrate and extract the features and finally achieve PolSAR classification.
2) A novel contrastive learning method is introduced and combined with ViT, which greatly reduces the redundancy of the baselines. Stop-gradient operation replaces the previous commonly used tools, including negative samples, memory bank, and large batch size, to realize a simple unsupervised network pretraining. The projection head is frozen to match the switch of backbone from CNN to ViT.
3) Experimental results on four PolSAR benchmark datasets show that both supervised and unsupervised

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ViT-based methods have better performance when inputting image patches. In addition, the comparison with hybrid models, as well as the research on initial input forms, illustrates the efficiency of the proposal.

The rest of this article is organized as follows. Section II presents a brief review of research background. The proposed framework is introduced in Section III. Experimental results and analyses are presented in Section IV. Conclusions and possible future directions are given in Section V.

II. Background

As an emerging technique, deep learning has made remarkable achievements in many fields, especially in computer vision and natural language processing. Its main manifestation is supervised learning, whose objective is to learn a model from labeled samples and infer new samples according to this model. A good candidate for such a model is undoubtedly CNN when the data to be processed are images. Compared with classical feature extractors, the unique of CNNs is the automated feature engineering process. In CNNs, feature extraction is mainly implemented by convolution, which can be defined as follows:

$$y=\sum_{c} x_{:,:,c} * w_{:,:,c} + b = \sum_{c}\left(\sum_{h}\sum_{w} x_{h,w,c}\cdot w_{h,w,c}\right) + b \tag{1}$$

where x and y denote input and output feature maps, respectively, h and w are the spatial dimensions, c is the channel dimension, and w and b are convolution kernel and bias, respectively, whose parameters are learnable. The values of w and b will be updated according to the optimization of objective function. Except for convolution, pooling is an important component to integrate information and reduce computational complexity, and the activation function is used to improve the nonlinear fitting ability of the model. Convolution, activation, and pooling are combined and stacked to form a hierarchical architecture, which enables CNNs to obtain hierarchical features. Obviously, the feature mapping of CNNs is not preset, but data-driven. The advantage is that it can fit the training set by minimizing the objective function to obtain good performance. However, a sufficient number of labeled samples to ensure the quality of training are the foundation. When this premise is not satisfied, the model will overfit the training data, resulting in low testing accuracy.

At present, the research of PolSAR representation learning and classification mainly follows the paradigm proposed in [21], which can be briefly seen from Fig. 1. Without loss of generality, it can be summarized as follows. First, PolSAR image data should be converted into an initial form, and all pixels with a small neighborhood around them are extracted to generate image patches. Based on the labeled pixel-centric PolSAR image patches, the training and testing sets can be divided as the input. Then, a backbone, usually convolutional architecture with multiple cascaded layers, is used to construct representations. Finally, a network head is added to implement image classification using the constructed representations. It is worth noting that the pipeline has a high degree of flexibility

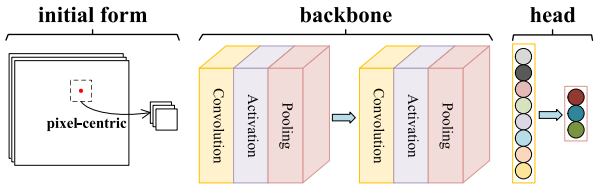

Fig. 1. Overview of CNN-based PolSAR classification paradigm.

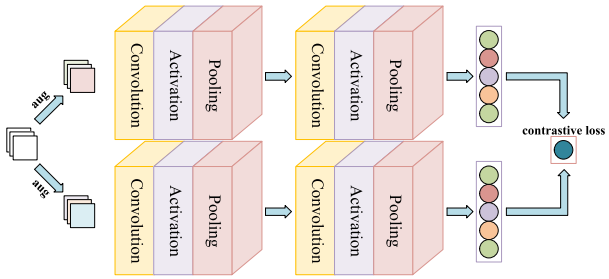

Fig. 2. Illustration of contrastive learning paradigm.

so that the front (initial form of PolSAR images), middle (backbone, CNN or others), and back can be replaced. For example, many initial forms and backbones have been introduced for better performance [22]–[24], and the output of the backbone can be used for classification [21] or detection [53] by customizing heads.

The objective function of supervised learning is a measure of similarity between annotations and predictions. Thus, annotation provides the driving force for model optimization and updating. However, in some real-world scenarios, it is very expensive or even impossible to obtain a large number of labeled samples, which limits the promotion seriously. Fortunately, the unsupervised approach has not been ignored [54]–[56] and is returning to the mainstream [57]. Recently, a long-established paradigm [58], i.e., contrastive learning, has been taken seriously and improved continuously.

As shown in Fig. 2, it transforms the input into two different augmentations by adding noise, cropping, flipping, or other ways [57]. Then, a dual-path convolutional architecture is used to process the two data augmentations, and the similarity of the products is measured via a contrastive loss function [59]. In order to accommodate unsupervised PolSAR representation learning, some pioneering works have been carried out [47], [49]. The existing results indicate that this appears to be a promising way to exploit the potential of contrastive learning in PolSAR images.

III. Methodology

In this section, the proposed ViT-based classification methods are introduced in detail. First, we review initial forms of PolSAR data, followed by detailed implementation of the proposal. The commonality between the supervised and unsupervised versions is that their feature extractions are performed by ViT.

A. PolSAR Data and Initial Forms

The research begins with the 2-D image data obtained from radar echo imaging. Each pixel of PolSAR images can be

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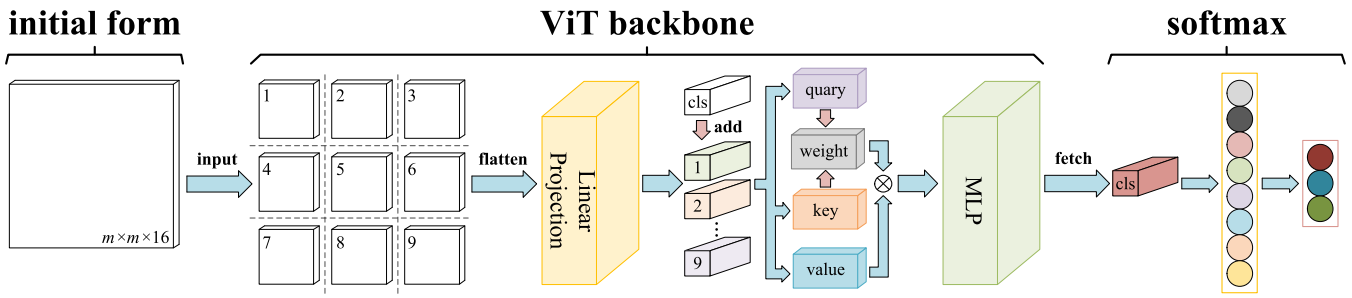

Fig. 3. Illustration of the supervised ViT-based PolSAR image classification. A labeled pixel-centric PolSAR image patch is chosen from the training set and partitioned into N^2 parts to become an image-generated sequence ($N = 3$ in the graph).

expressed by a complex scattering matrix

$$S = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{2}$$

where $S_{\mathrm{PQ}}(\mathrm{P}, \mathrm{Q} \in \{\mathrm{H}, \mathrm{V}\})$ represents the backscattering coefficient of the polarized electromagnetic wave in emitting Q direction and receiving P direction. H and V represent the horizontal and vertical polarizations, respectively. Based on the Pauli decomposition [7], the scattering vector can be defined as

$$\vec{k} = \frac{1}{\sqrt{2}}[S_{\mathrm{HH}} + S_{\mathrm{VV}}, S_{\mathrm{HH}} - S_{\mathrm{VV}}, 2S_{\mathrm{HV}}]'. \tag{3}$$

Then, the scattering matrix S can be transformed into the polarimetric coherency matrix T to describe the scattering properties of targets more clearly

$$T = \langle \vec{k}\vec{k}^H \rangle = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}. \tag{4}$$

Noting the fact that the polarimetric coherency matrix T is a Hermitian matrix, every element is a complex number except for the diagonal ones. Generally, the upper triangular elements $[T_{11}, T_{12}, T_{13}, T_{22}, T_{23}, T_{33}]$ can be taken as the initial form of PolSAR data.

Except for the polarimetric coherency matrix T, it is also possible to use polarimetric features as the input. Zhou *et al.* [21] proposed a 6-D real vector representation (RVR) as the input of CNN, which can be defined as

$$\mathrm{RVR}_1 = 10\log_{10}(T_{11} + T_{22} + T_{33}) \tag{5}$$

$$\mathrm{RVR}_2 = T_{22}/(T_{11} + T_{22} + T_{33}) \tag{6}$$

$$\mathrm{RVR}_3 = T_{33}/(T_{11} + T_{22} + T_{33}) \tag{7}$$

$$\mathrm{RVR}_4 = |\, T_{12} \,| \sqrt{T_{11} \cdot T_{22}} \tag{8}$$

$$\mathrm{RVR}_5 = |\, T_{13} \,| \sqrt{T_{11} \cdot T_{33}} \tag{9}$$

$$\mathrm{RVR}_6 = |\, T_{23} \,| \sqrt{T_{22} \cdot T_{33}}. \tag{10}$$

Besides, Chen *et al.* [22] proposed an adaptive initial form, which includes roll-invariant polarimetric features and rotation ones. Specifically, to formulate roll-invariant features, a coherency matrix should be parameterized based on the eigenvalue–eigenvector [15] as follows:

$$T = U \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} U^H. \tag{11}$$

Three polarimetric features, including the entropy H, mean alpha angle $\bar{\alpha}$, and anisotropy Ani, can be formulated by

$$H = -\sum_{n=1}^{3} P_n \log_3 P_n, \quad \bar{\alpha} = \sum_{n=1}^{3} P_n \alpha_n, \quad \mathrm{Ani} = \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3} \tag{12}$$

where $P_n = \lambda_n/(\lambda_1 + \lambda_2 + \lambda_3)$ and α is related to the matrix U. Another roll-invariant feature, i.e., total backscattering power SPAN, can be defined as

$$\mathrm{SPAN} = \langle |\, S_{\mathrm{HH}} \,|^2 \rangle + 2\langle |\, S_{\mathrm{HV}} \,|^2 \rangle + \langle |\, S_{\mathrm{VV}} \,|^2 \rangle. \tag{13}$$

According to [60], a 2-D rotation feature can be written as

$$\theta_{\mathrm{null_Re}}[T_{12}] = -\frac{1}{2}\mathrm{Angle}\{\mathrm{Re}[T_{13}] + j\mathrm{Re}[T_{12}]\} \tag{14}$$

$$\theta_{\mathrm{null_Im}}[T_{12}] = -\frac{1}{2}\mathrm{Angle}\{\mathrm{Im}[T_{13}] + j\mathrm{Im}[T_{12}]\} \tag{15}$$

where $\mathrm{Re}[\cdot]$ and $\mathrm{Im}[\cdot]$ denote the real and imaginary parts, respectively, and $\mathrm{Angle}\{\cdot\}$ is the operator to obtain the phase of complex numbers within the range of $[-\pi, \pi]$. Based on these works, Qin *et al.* [27] proposed a 16-D PolSAR initial form: $[T_{11}, T_{22}, T_{33}, \mathrm{Re}[T_{12}], \mathrm{Im}[T_{12}], \mathrm{Re}[T_{13}], \mathrm{Im}[T_{13}], \mathrm{Re}[T_{23}], \mathrm{Im}[T_{23}], \lambda_3, \bar{\alpha}, \mathrm{Ani}, \mathrm{RVR}_1, \mathrm{RVR}_4, \mathrm{RVR}_5, \mathrm{RVR}_6]$, which obtained satisfied results. High-dimensional data composed of various polarimetric features have been studied to achieve better decouplability [25], [26]. However, classical convolutional architectures may not be able to deal with the increase of input dimension leisurely, and variant models, such as squeeze-and-excitation module [61], are often required.

B. Supervised PolSAR Classification via ViT

An overview of the ViT-based PolSAR image classification is presented in Fig. 3, which illustrates the supervised version. Following the mainstream paradigm [21], pixel-centric PolSAR image patches are extracted, and the training set is prepared by random sampling from the labeled patches. Here, the 16-D PolSAR initial form [27] is utilized. Then, the patches are input into a ViT backbone to build representations. Finally, a softmax classifier with the cross-entropy loss function is used for supervised classification. Similar to convolutional architectures, ViT can also be reconstructed layer-by-layer, including data preprocessing layer, self-attention layer, and MLP layer.

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1) Data Preprocessing Layer: As shown in Fig. 3, an $m \times m \times 16$ PolSAR image patch is input into the model. Since ViT receives the 2-D input, this layer needs to reshape and map the 3-D image patch into a 2-D pixel embedding (corresponding to token embeddings in a standard transformer). First, the patch should be evenly divided into N^2 parts, and each part is expanded according to the spatial dimension to form a $((m/N) \cdot (m/N) \cdot 16)$-D vector. Stack all the vectors and the input will be reshaped from $\mathbb{R}^{m\times m\times 16}$ into $\mathbb{R}^{N^2\times((m/N)\cdot(m/N)\cdot 16)}$. At this point, an image-generated sentence is prepared where each element can be seen as the token in language.

Specifically, each partition of the original image patch will contain only one pixel when the value of N is equal to m. Then, a learnable matrix $W_{\mathrm{LP}}^{(1)} \in \mathbb{R}^{16\times d}$ is used for linear projection. We call the product of matrix multiplication pixel embeddings. The operation mentioned above is consistent with the fully connected layer of neural networks, except for the lack of nonlinear activation, which can be regarded as simple feature extraction.

After getting the pixel embeddings, a randomly initialized vector called class embedding is prepared. It is learnable and its dimension is consistent with that of the pixel embedding. Then, the two should be stacked to form a new pixel embedding with the size of $\mathbb{R}^{(1+m^2)\times d}$ as the input of next layer, where 1 denotes class embedding and m^2 is the old pixel embeddings. During the learning process, class embedding is used to integrate the information of the rest and finally represent the original PolSAR image patch. Different from mainstream ViT models [50], there is no position embedding in this work. Two factors can support this idea: On the one hand, there is no strict position relationship between pixels in the input PolSAR image patches. On the other hand, the existing methods may not be able to learn the real position information [62].

2) Self-Attention Layer: Similar to the role of convolution in CNNs, ViT features are mainly constructed through self-attention mechanism. Obviously, we need to get the importance of each pixel relative to others in order to capture the long-range interactions between them. This idea guides the implementation of self-attention, which aims to produce a weighted average of the embedded values. When the pixel embeddings are input, they should be linear mapped through a learnable matrix $W_{\mathrm{LP}}^{(2)} \in \mathbb{R}^{d\times 3d_h}$ into three variables with the same dimension: query q, key k, and value v. Then, information extraction and representation construction can be implemented through two steps, i.e., the acquisition and utilization of attention. First, query and key are used to generate a set of weight coefficients based on the pair-wise similarity between two elements of the sequence. Note the weight matrix as A, and its ith element can be written as

$$\begin{aligned} A_{i,:} &= \mathrm{metric}(q_i, k) = \mathrm{softmax}\left(\frac{q_i \cdot k}{\sqrt{d_h}}\right) \\ &= \mathrm{softmax}\left(\frac{1}{\sqrt{d_h}}\left(q_i k'_1, \ldots, q_i k'_i, \ldots, q_i k'_{N'}\right)\right) \end{aligned} \tag{16}$$

where $N' = 1 + m^2$ and $\mathrm{softmax}(x_i) = e^{x_i} / \sum_j e^{x_j}$. It can be seen from the definition that the similarity is obtained by calculating the dot product of q and k, which is related to the cosine similarity. A scaling coefficient $1/(d_h)^{1/2}$ is applied and softmax activation is used to map the output to $(0, 1)$ so as to generate weights. Equation (16) determines the importance of the ith position by measuring the correlation between q_i and k. Such a data-driven attention generation method often has a better performance than handcrafted methods [51]. Then, the output SA can be obtained by

$$\mathrm{SA} = Av = \begin{bmatrix} A_{1,1}v_1 + \cdots + A_{1,N'}v_{N'} \\ \vdots \\ A_{N',1}v_1 + \cdots + A_{N',N'}v_{N'} \end{bmatrix}_{N'\times d_h} \tag{17}$$

where the ith element of SA is a weighted sum of v and the importance of the ith position. This operation releases the attention obtained from (16) and constructs new representations based on the mapped pixel embeddings. Similar to the concept of the number of kernels in convolution, self-attention can also be calculated multiple times to improve the model capacity, which is called multihead self-attention. Times to repeat (16) and (17) are equal to the number of heads.

3) MLP Layer: In this section, two fully connected (FC) layers and a GeLU activation [44] will be used to integrate the obtained information. FC layers with nonlinear activation function have been proved to increase the model capacity, which is used here to make up for the small number of model parameters in the self-attention layer of ViT. As shown in Fig. 3, the first row of the output, i.e., the class embedding, is taken out as the representation and the rest m^2 dimensions are omitted. The image classification is implemented by a softmax classifier, which is added at the end of the model.

C. Unsupervised PolSAR Representation Learning and Classification via ViT

In our previous work [47], the potential of unsupervised PolSAR representation learning has been explored, where a momentum contrast-based contrastive learning method with a customized diversity stimulation mechanism is proposed. However, the algorithm contains multiple stages, so it has relatively high complexity and redundancy. This is not friendly for the implementation, reproduction, and future improvement of the algorithm. To handle this problem, in this work, a simple strategy [63] is introduced to greatly reduce the difficulty of algorithm implementation. On this foundation, unsupervised version of the ViT-based representation learning framework is developed, as shown in Fig. 4.

First, the input should be randomly augmented into two parts. In this work, the augmentation includes *resized crop* with the ratio of [0.4, 1.0], *horizontal flip*, and *Gaussian blur* with $\sigma \in [0.1, 1.0]$. Note the two augmentations as x_1 and x_2, and they should be processed by a ViT-based backbone $f(\cdot)$, two MLP-based projection, and prediction heads $g(\cdot)$ and $h(\cdot)$. The similarity between the products of the two paths can be measured according to the following formula:

$$\mathrm{Similarity}(x_1, x_2) = -\frac{h(g(f(x_1)))}{\|h(g(f(x_1)))\|_2} \cdot \frac{g(f(x_2))}{\|g(f(x_2))\|_2}. \tag{18}$$

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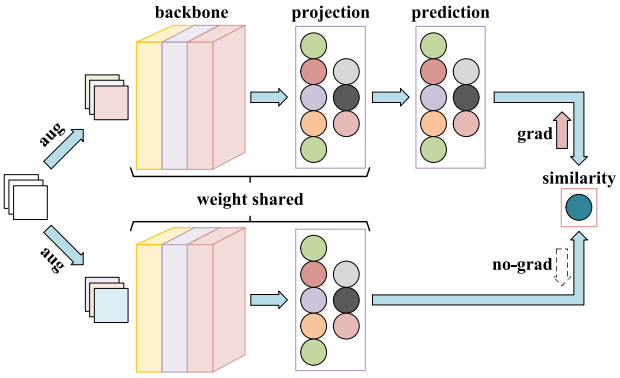

Fig. 4. Illustration of the unsupervised ViT-based PolSAR image representation learning. The backbone is a ViT architecture, and the projection and prediction heads are two MLP layers. Gradient backpropagation only occurs in the upper half of the two paths in the figure.

Compared with the widely used contrastive loss function [57], (18) is more easy to implement because the temperature hyperparameter τ that needs to be carefully adjusted is omitted. Similar to [64], (18) is symmetrized to form the following loss function:

$$\mathcal{L} = \frac{1}{2}\text{Similarity}(x_1, x_2) + \frac{1}{2}\text{Similarity}(x_2, x_1) \qquad (19)$$

where stop-gradient operation is applied to the second variable. Since stop gradient only occurs on one of the two paths of the model and the other path can still implement error backpropagation, thus, this operation will not affect the model updating. Since the ViT-based backbone is used instead of a convolutional one, the training process will become unstable. To handle this problem, the model parameters of projection head are frozen to the initial values during the training [62]. Different from the mainstream methods, this is a very simple strategy and it does not contain the complex tools needed for unsupervised learning, including temperature hyperparameter [65], negative samples [66], memory bank [67], and large batch size [57].

A large number of unlabeled PolSAR data should be prepared to train the backbone and prediction head. Then, the pretrained backbone should be taken out and its output is considered as high-level representations, which can be used for supervised or unsupervised PolSAR classification [47]. In this work, the pretrained model should be evaluated by the linear evaluation protocol [68] to verify the effectiveness of unsupervised learning. It means that the backbone should be frozen after pretraining, and a softmax classifier performs classification based on the representations constructed by the pretrained backbone. The whole process is shown in Algorithm 1.

Algorithm 1 Unsupervised ViT-Based PolSAR Image Representation Learning and Classification

Begin
Prepare unlabeled and labeled pixel-centric PolSAR image patches $\Theta_1 = \{\mathbf{z}_i\}_{i=1}^{M_1}$, $\Theta_2 = \{\mathbf{z}_i\}_{i=1}^{M_2}$.
Require: Pre-training
Set the hyperparameters of $f(\cdot)$, $g(\cdot)$ and $h(\cdot)$.
Prepare the number of epoch $\mathcal{I}_1$, steps B_1. Initialize the parameters of $f(\cdot)$, $g(\cdot)$ and $h(\cdot)$. Define optimizer and learning rate.
for iteration in $\mathcal{I}_1$: **do**
  for mini-batch in B_1: **do**
    Random augment the input samples.
    Network forward propagation.
    Calculate the similarity-based loss (19).
    Loss backpropagation on the variable of interest.
  end for
end for
return Pre-trained backbone $f^*(\cdot)$.
Ensure: Linear evaluation
Divide Θ_2 into training, validation and testing sets.
Prepare the number of epoch $\mathcal{I}_2$, steps B_2. Set the backbone to $f^*(\cdot)$ and freeze. Initialize the parameters of softmax classifier. Define optimizer and learning rate.
for iteration in $\mathcal{I}_2$: **do**
  for mini-batch in B_2: **do**
    Network forward propagation.
    Calculate the cross-entropy loss.
    Classifier loss backpropagation.
  end for
end for
return Classifier with the highest validation accuracy.
Feed testing samples into $f^*(\cdot)$ and the best classifier.
return Test accuracy.

IV. Experimental Results

A. Datasets Description

We employ four widely used PolSAR datasets in the experiments. The first three datasets are acquired by airborne PolSAR, and ALOS2 is obtained by spaceborne PolSAR. Figs. 5–8 show their Pauli maps and ground truth maps.

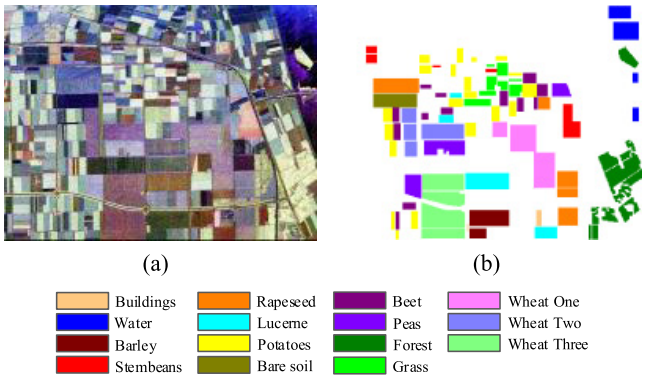

Fig. 5. AIRSAR Flevoland dataset and its color code. (a) Pauli RGB map. (b) Ground truth map.

1) AIRSAR Flevoland: Fig. 5 shows an L-band, full-polarimetric image of the agricultural region of the Netherlands. The size of this image is 750 × 1024. There are 15 kinds of ground objects in this dataset.

2) ESAR Oberpfaffenhofen: An L-band, full-polarimetric image of Oberpfaffenhofen, Germany, is 1200 × 1300 scene

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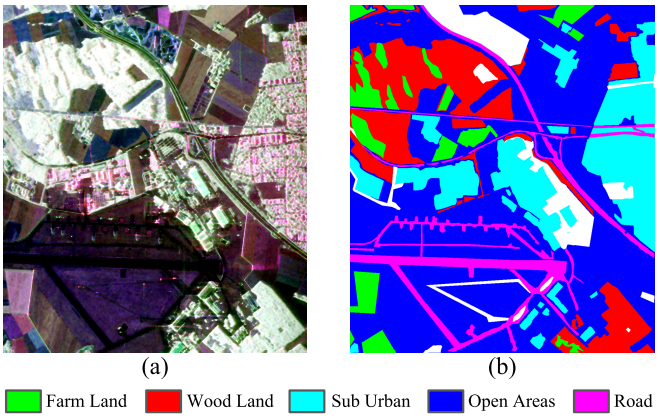

Fig. 6. ESAR Oberpfaffenhofen dataset and its color code. (a) Pauli RGB map. (b) Ground truth map.

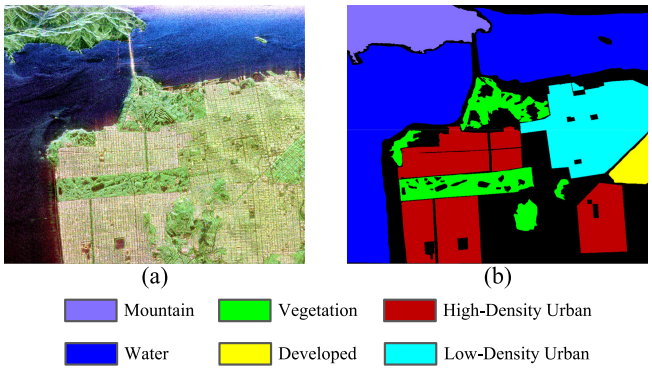

Fig. 7. AIRSAR San Francisco dataset and its color code. (a) Pauli RGB map. (b) Ground truth map.

size. Its Pauli color-coded image and ground truth map can be seen in Fig. 6. Each pixel in the map can be divided into five categories [69].

3) AIRSAR San Francisco: This is an L-band, full-polarimetric image of San Francisco, as shown in Fig. 7. The size of this image is 900 × 1024. Each pixel of this image can be classified into six categories [70].

4) ALOS2 San Francisco: This is an L-band, full-polarimetric image of San Francisco, as shown in Fig. 8. The size of this image is 5056 × 2784. Each pixel of this image can be classified into six categories [70].

B. Experimental Setup

The experiments are carried out on a personal computer with a NVIDIA GeForce GTX 1060 GPU with 6-GB memory. In the following, we introduce the implementation details of supervised and unsupervised experiments.

1) Supervised Experiment Settings: For supervised ViT-based representation learning, the neighborhood around labeled pixels is set to be 14 × 14. Therefore, the value of input spatial size m is 15 and the number of pixel embeddings N' is 226. The hyperparameters of ViT are set as follows. The dimension of linear projection d is set to be 64. The number of times that SA is calculated, i.e., heads, is set to be 4 and the dimension of each head d_h is 16. The output of multihead self-attention should be linearly mapped and

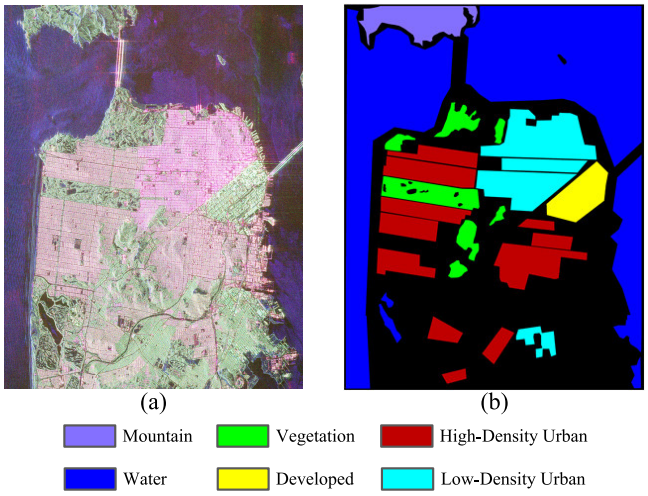

Fig. 8. ALOS2 San Francisco dataset and its color code. (a) Pauli RGB map. (b) Ground truth map.

connected with its input through a residual connection. The number of hidden nodes in the FC layers of MLP is 128. Residual connection is also used in the MLP layer. Before each layer of ViT, layer normalization is used to speed up the convergence [71].

According to the annotation given in Figs. 5–8, there are 184 592, 1423 542, 649 001, and 7986 515 labeled samples in Flevoland, Oberpfaffenhofen, and two San Francisco datasets, respectively. For the experiments in different datasets, we randomly select 300 samples from each category as the training sets. In the remaining samples, 600 (Flevoland) or 2000 (Oberpfaffenhofen and San Francisco) samples per category are randomly selected to form the validation sets. Except for the training sets, all labeled samples are used for testing in order to measure the performance.

In order to evaluate the effectiveness of the proposal, several supervised methods are performed and tested. On the one hand, in the experiments, Wishart [72] and radial basis function kernel-based support vector machine (RBF-SVM) [17] are used as nondeep learning methods for comparison. On the other hand, some recently proposed deep learning-based methods are chosen for comparison, including CNN [21], CV-CNN [29], 3-D-CNN [28], PSENet [27], and polarimetric-feature-driven CNN (SF-CNN) [22]. In the experiments, all the models are trained for 150 epoch with the batch size of 256. The adam optimizer [73] is utilized to train these models and the initial learning rate is set to 0.001.

2) Unsupervised Experiment Settings: Data format of unsupervised experiments roughly follows the settings of supervised, and the hyperparameters are slightly different from the previous settings. For unsupervised version, inspired by [62], we set the spatial size of input as 7 and increase the depth of the backbone to twice the previous. Such setting increases the amount of model parameters, and the dimension of all linear projection operations is set to half of the previous so as to reduce the computational complexity. In addition, the layer normalization of MLP layers is replaced by batch normalization [74]. The projection and prediction heads contain two FC layers with ReLU activation [75] and batch normalization.

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TABLE I
EXPERIMENTAL RESULTS (%) OF SUPERVISED METHODS ON THE AIRSAR FLEVOLAND DATASET

Method	Wishart	RBF-SVM	CNN	CV-CNN	3D-CNN	PSENet	SF-CNN	ViT
Buildings	68.02	87.54	99.48	99.90	99.38	99.69	100.00	100.00
Rapeseed	50.16	85.04	82.94	92.95	87.77	86.48	95.12	98.90
Beet	91.86	90.91	96.68	98.06	95.36	95.26	97.24	97.84
Stembeans	93.04	88.21	96.95	98.83	98.05	97.56	99.53	99.88
Peas	87.06	94.38	97.28	97.01	96.94	94.49	98.03	99.32
Forest	72.94	85.19	96.14	98.28	93.35	93.74	97.25	98.76
Lucerne	90.82	89.10	92.56	94.08	93.76	92.09	95.90	98.95
Potatoes	58.83	81.50	94.03	92.15	89.76	93.50	91.10	97.52
Bare soil	94.46	94.72	97.53	98.72	97.76	97.94	99.84	99.95
Grass	65.36	66.01	77.42	92.90	91.98	88.06	89.10	98.01
Barley	94.24	94.95	96.84	96.35	96.54	96.84	98.02	99.66
Water	52.95	97.67	98.49	99.03	98.12	100.00	100.00	100.00
Wheat One	89.76	79.11	93.74	96.26	95.34	91.38	94.38	98.66
Wheat Two	84.41	78.57	82.84	89.97	87.03	93.38	92.82	99.17
Wheat Three	82.44	86.73	93.69	96.75	93.74	91.04	94.96	99.40
AA	78.42	86.64	93.11	96.08	94.33	94.10	96.22	**99.07**
Kappa	75.15	84.50	91.84	95.18	92.68	92.35	95.08	**98.79**
OA	77.15	85.77	92.52	95.58	93.28	92.97	95.48	**98.89**

TABLE II
EXPERIMENTAL RESULTS (%) OF SUPERVISED METHODS ON THE ESAR OBERPFAFFENHOFEN DATASET

Method	Wishart	RBF-SVM	CNN	CV-CNN	3D-CNN	PSENet	SF-CNN	ViT
Sub Urban	31.90	63.18	75.48	72.33	75.74	82.49	82.63	88.72
Farm Land	44.29	61.11	73.23	83.03	86.10	84.79	85.49	87.45
Road	88.08	88.24	64.38	80.31	80.10	78.50	83.67	78.62
Open Areas	44.61	53.98	74.46	69.98	70.33	72.46	72.41	70.68
Wood Land	80.07	81.82	87.45	84.68	85.94	83.25	81.47	87.48
AA	57.79	69.67	75.00	78.07	79.64	80.30	81.14	**82.59**
Kappa	37.31	50.61	64.71	63.76	65.32	67.27	67.63	**68.87**
OA	51.37	63.01	75.69	74.21	75.33	77.02	77.22	**77.93**

The number of hidden nodes in the projection head is 64 and 32 and that in the prediction head is 16 and 32.

Unsupervised model pretraining requires the support of a large number of samples. In this work, more than 60 000 samples are selected from each PolSAR dataset for training. Since performance evaluation is not required in the stage of pretraining, there are no validation and testing sets. When linear evaluation needs to be carried out, the previous training, validation, and testing sets will be reused.

To verify the capacity of ViT as a backbone, in this section, the same model with a CNN backbone is constructed for comparison. In the experiments, pretraining is implemented for 150 epoch. The batch size of these experiments is set to 256. A stochastic gradient descent (SGD) optimizer with the momentum of 0.9 and weight decay of 0.0001 is used for model pretraining. The initial learning rate is set to 0.05, followed by cosine schedule. For linear evaluation, the classifier is trained for 50 epoch with the Adam optimizer whose learning rate is set to 0.1. Weight decay or similar regularization tricks are not enabled.

There are three kinds of evaluation criteria including average accuracy (AA), Kappa coefficients (Kappa), and overall accuracy (OA) in the experiments. Among them, OA and AA can measure the general performance of algorithms and Kappa is more important in the case of imbalanced data classification.

C. Classification Results of PolSAR Datasets

In this section, we introduce the supervised and unsupervised classification results on four datasets. In addition, the comparative results with hybrid models and the influence of initial input forms will also be fully discussed.

1) Supervised Results: The supervised experimental results on involved four PolSAR benchmark datasets are shown in Tables I–IV. The whole map classification results are shown in Figs 9–12. Generally speaking, the classification ability of ViT is significantly stronger than that of comparing methods. The performance of shallow classifiers, i.e., Wishart and RBF-SVM, is seriously limited by model capabilities. Therefore, they do not show the competitiveness compared with the deep learning-based methods. The performance of four CNN variant methods is improved compared with the basic method. Compared with the CNN baseline, the classification accuracy of the proposed framework has been improved by about 6% at most, which indicates that the performance can be greatly improved. Looking at those with better performance in the

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TABLE III
EXPERIMENTAL RESULTS (%) OF SUPERVISED METHODS ON THE AIRSAR SAN FRANCISCO DATASET

Method	Wishart	RBF-SVM	CNN	CV-CNN	3D-CNN	PSENet	SF-CNN	ViT
Mountain	67.08	82.80	88.23	89.74	93.88	92.45	93.52	96.47
Water	99.95	99.02	99.56	99.14	99.52	99.87	99.93	99.52
Vegetation	52.32	60.54	73.55	80.17	84.62	79.19	84.83	93.47
High-Density Urban	49.44	73.19	87.76	84.91	94.32	84.99	88.77	94.38
Low-Density Urban	54.08	72.48	86.46	87.60	90.64	88.66	85.83	96.02
Developed	61.30	67.03	83.02	81.85	92.86	81.70	90.35	95.86
AA	64.03	75.84	86.43	87.23	92.64	87.81	90.54	**95.95**
Kappa	65.27	77.98	88.06	88.29	93.20	89.15	90.80	**95.88**
OA	74.40	83.96	91.33	91.47	95.07	92.11	93.32	**97.01**

TABLE IV
EXPERIMENTAL RESULTS (%) OF SUPERVISED METHODS ON THE ALOS2 SAN FRANCISCO DATASET

Method	Wishart	RBF-SVM	CNN	CV-CNN	3D-CNN	PSENet	SF-CNN	ViT
Mountain	76.04	57.26	61.04	69.10	69.91	70.46	66.97	78.43
Water	95.23	98.87	99.06	99.68	99.52	99.46	99.44	99.50
Vegetation	32.26	37.66	58.95	65.43	67.21	58.13	67.19	68.50
High-Density Urban	52.88	57.54	79.80	84.46	86.49	78.78	88.00	88.94
Low-Density Urban	54.65	68.29	89.70	87.06	89.72	92.23	88.66	94.99
Developed	37.09	48.71	77.88	76.74	84.70	83.47	85.77	92.90
AA	58.02	61.39	77.74	80.41	82.93	80.42	82.67	**87.21**
Kappa	63.70	70.14	84.04	86.42	87.97	85.48	87.94	**90.76**
OA	77.18	81.58	90.19	91.68	92.63	91.09	92.61	**94.34**

variants, the proposal also has a certain lead. The experimental results on each dataset are analyzed as follows.

For the experiments on the AIRSAR Flevoland dataset, compared with the ground truth in Fig. 5, it can be seen that the proposal has better classification performance. The whole map classification results are shown in Fig. 9 and the numerical results are listed in Table I. From the results in Fig. 9, it is obvious that the land cover classification map of ViT is more pure in each rectangle partition than comparing methods. For the results in Table I, the proposal achieves the highest classification precision in 14 out of 15 categories, except that it lags behind CV-CNN by 0.22% in the beet category. The increase in almost all categories brings significant improvements in AA, Kappa, and OA, and there are 2.85%, 3.61%, and 3.31% increases of criteria compared with the best of comparing methods. This may mean that when processing the data with a relatively high number of categories, the global features obtained by ViT will be more effective and the classification results will be satisfactory.

Fig. 10 and Table II summarize the graphical and numerical classification results on the ESAR Oberpfaffenhofen dataset. Similar to previous studies, confusion between man-made areas and woodland areas is a common problem. As can be seen from the cyan area of classification maps, the confusion given by the ViT-based method is the least of all methods. Moreover, the open areas represented by blue contain a variety of crops, which can be easily confused with the farmland represented by green. The ViT-based method handles this problem well. Table II shows the results of involved methods on the testing set, from which one can see that global information does enhance the quality of representation and classification. There are 1.45%, 1.24%, and 0.71% accuracy increases for AA, Kappa, and OA compared with the proposal and the best of comparing methods, respectively.

The whole map testing results on the AIRSAR San Francisco dataset are shown in Fig. 11. It can be clearly seen that the proposal has better classification performance. Especially for the man-made areas in the image, including high- (red) and low-density urban (cyan), the ViT-based method greatly reduces the probability of misclassification. In addition, the proposed method can distinguish vegetation and urban more accurately, which is not well done by other methods. The accuracy of each method on the testing set is shown in Table III to quantitatively compare the performance. It can be seen that the ViT-based representations obtain the best results on the involved criteria. Compared with the CNN baseline, the AA, Kappa, and OA are improved by 9.52%, 7.82%, and 5.68%, respectively. Besides, there are 3.31%, 2.68%, and 1.94% increases of criteria compared with the best of comparing methods.

For the other San Francisco dataset, due to the difference between the shooting time and the sensor, the acquired data have changed significantly, so the classification result has also changed. However, the relative merits of the involved methods are similar to previous experiments. As shown in Fig. 12, the classification map of ViT has less wrong points in urban areas. Also, developed areas (yellow) on the map are also less misclassified as vegetation (green) by ViT. The quantitative experimental results are reflected in Table IV, where the performance of the proposed method is significantly improved.

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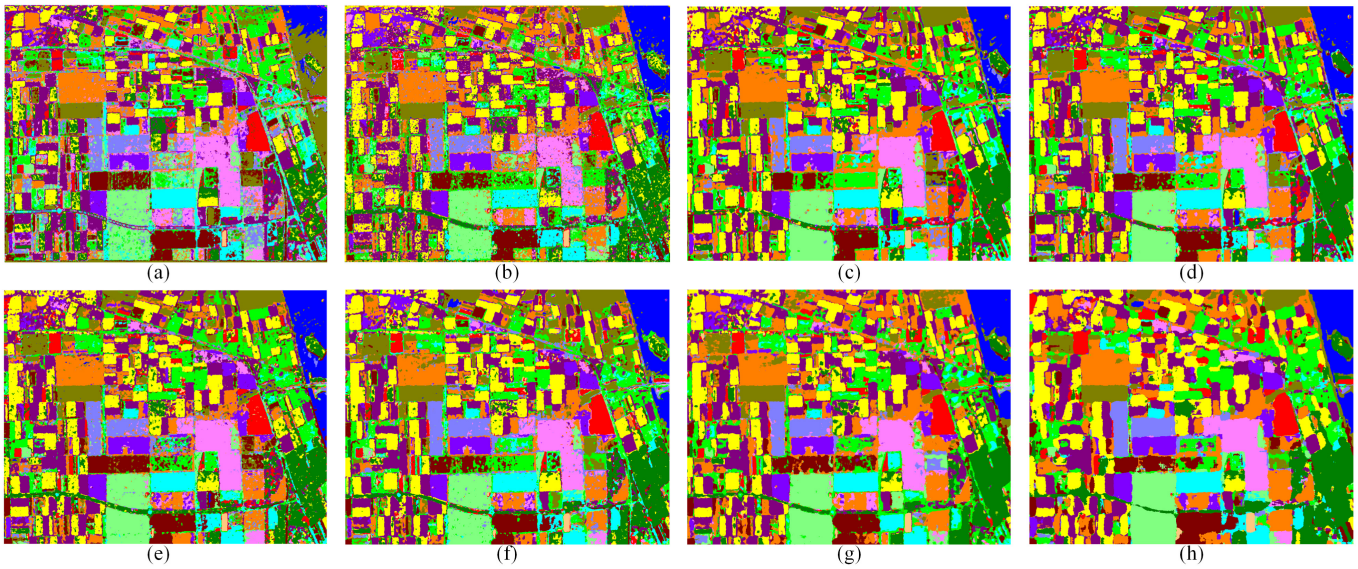

Fig. 9. Supervised classification results of the whole map on AIRSAR Flevoland dataset. (a) Result of Wishart. (b) Result of RBF-SVM. (c) Result of CNN. (d) Result of CV-CNN. (e) Result of 3D-CNN. (f) Result of PSENet. (g) Result of SF-CNN. (h) Result of ViT.

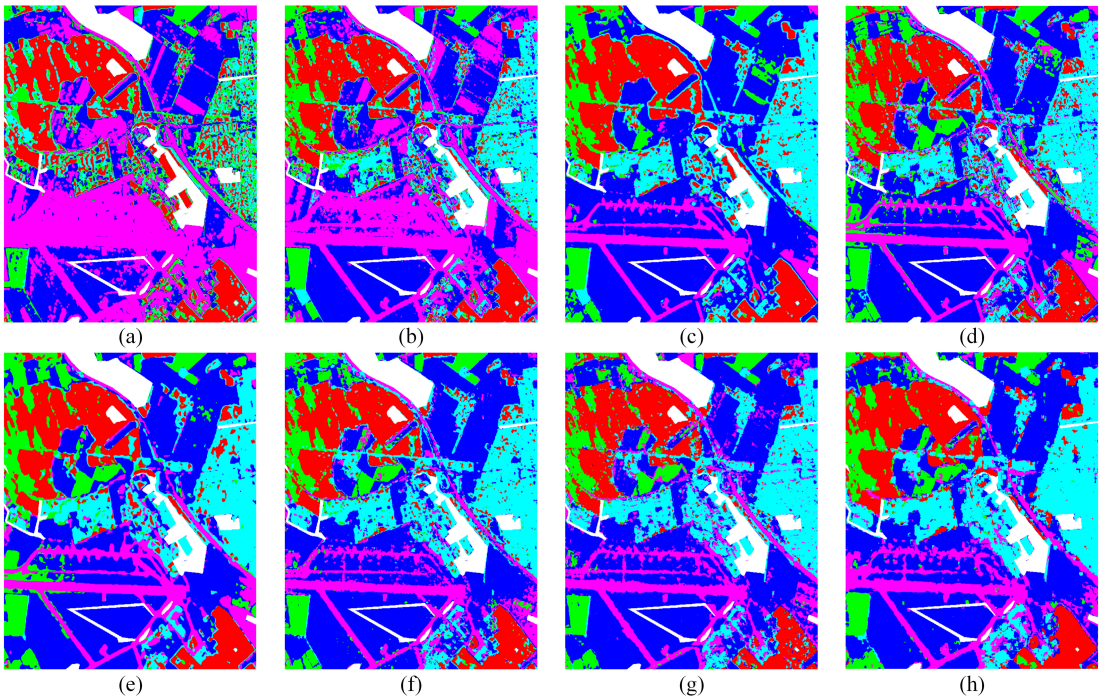

Fig. 10. Supervised classification results of the whole map on ESAR Oberpfaffenhofen dataset. (a) Result of Wishart. (b) Result of RBF-SVM. (c) Result of CNN. (d) Result of CV-CNN. (e) Result of 3D-CNN. (f) Result of PSENet. (g) Result of SF-CNN. (h) Result of ViT.

Compared with other depth learning-based methods, ViT is more effective in identifying mountain and developed areas. Compared with the CNN baseline, ViT improves three criteria by 9.47%, 6.72%, and 4.15%. In addition to the proposed method, 3D-CNN achieves the best classification performance, which is 4.28%, 2.79%, and 1.71% less AA, Kappa, and OA than ViT, respectively.

In summary, the conclusions presented by the experimental results are consistent with the theoretical analysis, which indicates that for the processing of pixel-centric PolSAR image patches, the representation obtained by ViT-based backbone does have superior quality as well as classification performance compared to that obtained by convolutional architectures. In terms of accuracy improvement, the proposed method has the largest improvement on two San Francisco datasets, followed by the Flevoland dataset, and the smallest improvement on the Oberpfaffenhofen dataset. For the supervised classification task, the proposed method can achieve good performance on both airborne and spaceborne PolSAR data.

2) Unsupervised Results: In order to test the capacity of ViT in unsupervised tasks, based on the experiment settings described previously, unsupervised experiments are carried out on four involved PolSAR benchmark datasets.

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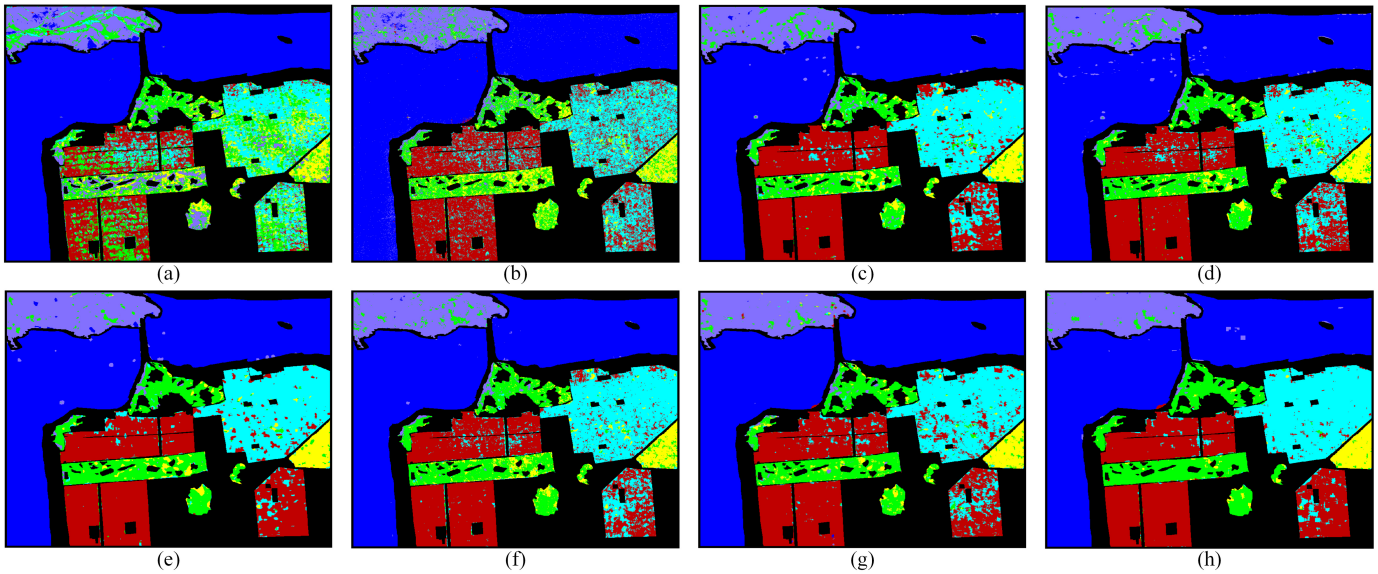

Fig. 11. Supervised classification results of the whole map on AIRSAR San Francisco dataset. (a) Result of Wishart. (b) Result of RBF-SVM. (c) Result of CNN. (d) Result of CV-CNN. (e) Result of 3D-CNN. (f) Result of PSENet. (g) Result of SF-CNN. (h) Result of ViT.

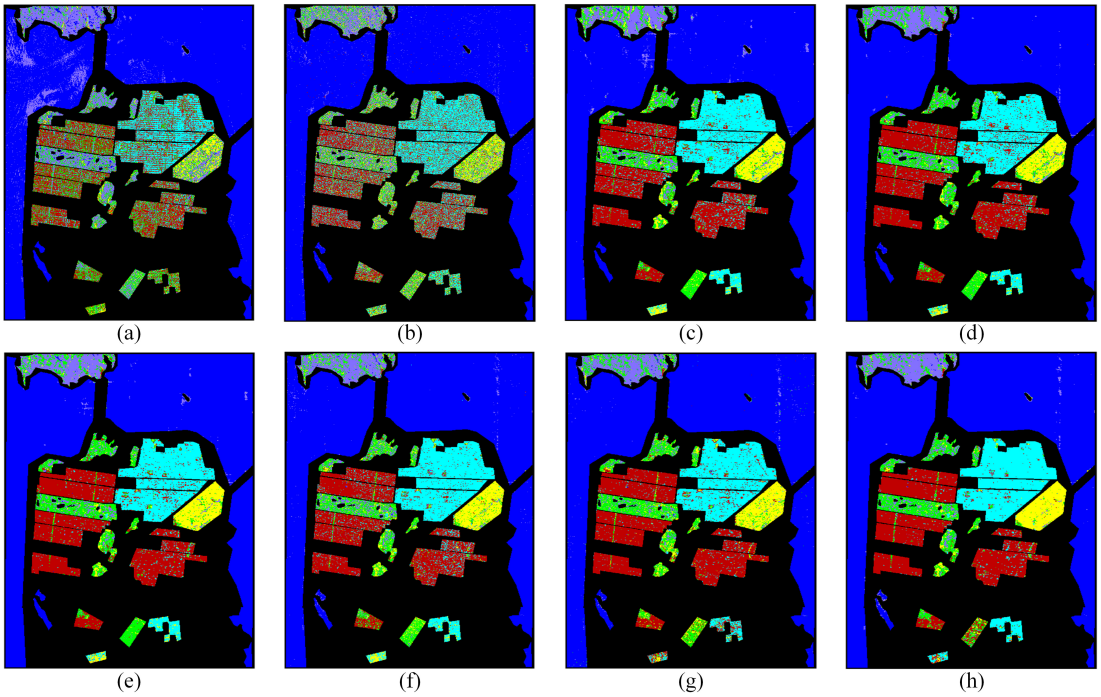

Fig. 12. Supervised classification results of the whole map on the ALOS2 San Francisco dataset. (a) Result of Wishart. (b) Result of RBF-SVM. (c) Result of CNN. (d) Result of CV-CNN. (e) Result of 3D-CNN. (f) Result of PSENet. (g) Result of SF-CNN. (h) Result of ViT.

The experimental results of the proposed unsupervised learning framework with two different backbones, i.e., CNN and ViT, are shown in Tables V–VII.

It can be seen from the results that the adjustment of backbone brings a certain performance improvement. In the experiments on four datasets, the minimum is 2.91%, 1.44%, and 0.96% leading of AA, Kappa, and OA on the AIRSAR San Francisco dataset, respectively, and $3\% - -5\%$ increases of AA, Kappa, and OA can be obtained on the other three datasets. Compared with CNN, the better linear evaluation performance confirms that the representation obtained by ViT in an unsupervised manner is of better quality for PolSAR image classification.

From another perspective, by comparing with the results in Tables I–IV, it can be seen that the performance of the proposed unsupervised method with linear evaluation protocol is satisfactory. The accuracy of ViT-based unsupervised model has a significant improvement compared with Wishart and RBF-SVM. Also, the accuracy of ViT-based unsupervised method is slightly behind that of supervised CNN on the three datasets except for ALOS2.

3) Comparison With Hybrid Models: The latest research is dedicated to giving ViT the ability to learn hierarchical representation like CNNs [76]. This is because, for the input image with a large size, local information is also important. It is simple to build a hybrid model by combining convolution

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TABLE V
UNSUPERVISED RESULTS (%) ON THE AIRSAR FLEVOLAND DATASET

Method	CNN	ViT
Buildings	90.34	95.53
Rapeseed	84.17	85.29
Beet	90.94	90.70
Stembeans	93.75	92.50
Peas	94.16	93.33
Forest	93.59	96.12
Lucerne	78.61	83.52
Potatoes	90.14	94.28
Bare soil	91.24	99.08
Grass	63.54	88.94
Barley	91.51	90.54
Water	73.29	83.11
Wheat one	86.70	88.67
Wheat two	90.38	91.75
Wheat three	87.79	87.04
AA	86.68	**90.69**
Kappa	86.04	**89.35**
OA	87.19	**90.21**

TABLE VI
UNSUPERVISED RESULTS (%) ON THE ESAR OBERPFAFFENHOFEN DATASET

Method	CNN	ViT
Sub Urban	48.22	68.69
Farm Land	57.04	46.17
Road	63.59	73.08
Open Areas	77.53	74.37
Wood Land	76.83	85.29
AA	64.64	**69.52**
Kappa	56.51	**61.40**
OA	70.06	**73.30**

and self-attention layers to achieve hierarchical feature extraction [77]. Considering that the size of pixel-centric PolSAR image patches is relatively small, the importance of local information may be weakened, and the performance of the hybrid model may not be as good as the standard ViT.

Comparative experiments with the hybrid model are carried out in order to confirm the above analysis and prove the advantages of the proposal. The results are shown in Table VIII, where the CNN-based method that performs relatively well in previous experiments, i.e., SF-CNN, is selected to be combined with ViT. There are three convolution layers in SF-CNN, and they should be extracted and combined to form hybrid models for comparisons. For example, select the first convolution layer of SF-CNN with the nonlinear activation and stack it with ViT form a hybrid model, called hybrid-layer-1 (H-L1). Similarly, the remaining two models (H-L2 and H-L3) can be constructed.

From the results, it can be seen that the proposal obtains the best accuracy without convolution operations. Moreover, as the number of convolution layers increases, the classification performance of hybrid models shows a clear downward trend. Noting the fact that the more times of convolution, the more attention is paid to local information, and the effect is negative. Therefore, the results support our motivation. The model directly builds the long-range interactions, which is more efficient than hierarchy architectures.

TABLE VII
UNSUPERVISED RESULTS (%) ON TWO SAN FRANCISCO DATASETS

	AIRSAR		ALOS2	
Method	CNN	ViT	CNN	ViT
Mountain	83.85	97.12	69.79	75.56
Water	100.00	98.58	97.90	99.42
Vegetation	59.94	54.95	43.58	33.75
High-Density Urban	80.67	82.61	84.46	75.36
Low-Density Urban	84.35	86.70	53.62	77.11
Developed	78.80	85.12	66.26	69.89
AA	81.27	**84.18**	69.27	**71.85**
Kappa	83.38	**84.82**	75.85	**79.02**
OA	87.95	**88.91**	85.12	**87.09**

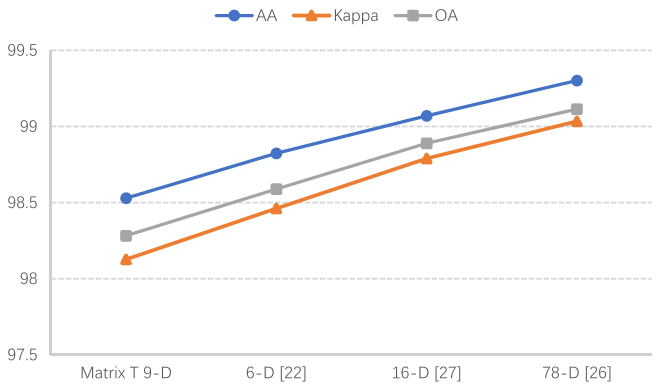

Fig. 13. Performance comparison of four initial input forms.

4) Influence of Initial Forms: Many tailored convolutional architectures [25]–[27] have been proposed to effectively use the information in different initial forms. Although these works have improved accuracy, they are also accompanied by the problems of insufficient adaptability and difficulty in reproduction. If the model can handle various initial input forms without making changes, it will be more friendly to applications. In this section, we test the sensitivity of ViT to the initial form of PolSAR data. There are totally four kinds of initial forms that are involved in the experiment, including 9-D coherency matrix T [29], the 6-D polarimetric feature used in SF-CNN [22], the 16-D feature in PSENet [27], and the 78-D feature used in [26]. Backbones used in the experiment are exactly the same, except for the dimension of input layers. Without loss of generality, the experiment is carried out on the Flevoland dataset, and the supervised classification results with different kinds of initial input forms can be seen from Fig. 13.

The experimental results show that the models using four different initial input forms all achieve good classification results, and the performance difference between them is relatively small. As the initial form becomes more and more compact, each criterion has a slight increase within 1%.

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TABLE VIII
EXPERIMENTAL RESULTS (%) OF HYBRID MODEL ON FOUR DATASETS

	AIRSAR Flevoland			ESAR Oberpfaffenhofen			AIRSAR San Francisco			ALOS2 San Francisco		
Criterion	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA
H-L1	97.28	96.74	97.01	92.43	92.25	94.38	84.46	88.39	92.87	80.59	65.81	75.64
H-L2	96.86	96.05	96.37	87.82	88.54	91.67	82.72	87.76	92.51	79.05	64.81	74.97
H-L3	95.11	94.20	94.68	86.24	86.95	90.52	81.16	86.83	91.94	79.24	63.43	73.71
ViT	99.07	98.79	98.89	95.95	95.88	97.01	87.21	90.76	94.34	82.59	68.87	77.93

Therefore, it can be concluded that ViT is robust to the initial forms for PolSAR classification task.

V. CONCLUSION

In this article, the capacity of language model for PolSAR image classification is explored. A ViT-based representation learning framework is proposed for constructing representations and implementing classification in both supervised and unsupervised manners. According to the characteristics of input PolSAR image patches, we focus on building the long-range interactions of pixels directly via self-attention mechanism instead of constructing convolution-based hierarchy architectures, which brings strong global information in the obtained representations. Since the locality of the convolutional architectures is improved, ViT can be expected to have good performance in pixel-centric PolSAR image patches classification. Besides, a contrastive learning method with low complexity and redundancy is introduced and combined with the ViT backbone to form an unsupervised version of the framework. Compared with the previous unsupervised data-driven methods, the proposal has stronger feature extraction ability, simplifies the learning process, and reduces the difficulty of implementation.

Like other deep learning-based classification methods, the supervised vision of the proposed framework can be used for PolSAR classification in an end-to-end manner. Experimental results on both airborne and spaceborne PolSAR datasets verify that the proposal has a large degree of improvement compared with other deep learning-based methods. This is largely due to the self-attention mechanism, which replaces the convolution in CNNs for feature extraction and improves the globality of features. In addition, our experiments also prove that ViT is robust to the initial form of PolSAR data, which means good flexibility in applications.

When switching to unsupervised learning, the proposed method uses similarity measures to train the network without manual annotations. Data-driven deep PolSAR representations can be constructed with the pretrained ViT models. In this work, we use a linear evaluation protocol to evaluate the results of unsupervised learning, as well as produce classification results. Compared with conventional methods, the unsupervised version is simple and easy to reproduce, which is beneficial to application and improvement. Numerical experiments on four datasets exhibit the validity of the proposed method.

This work covers supervised and unsupervised learning in deep PolSAR image classification domain. The pretrained network model can provide useful features for distinguishing different land cover types. Therefore, it can be used in both supervised and unsupervised ways. The satisfactory performance on these two issues comprehensively shows that ViT is effective in the PolSAR image classification. However, it is difficult to implement the self-attention mechanism on large-size input because of its quadratic computation complexity to input size. In future work, we will improve the supervised ViT and control the model complexity so as to adapt the task of dense prediction. At the same time, exploring the effectiveness of the unsupervised version in other PolSAR applications, including target recognition, detection, and change detection, is also the direction of our future work.

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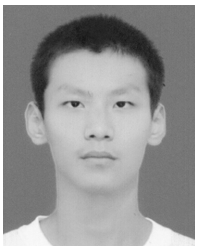

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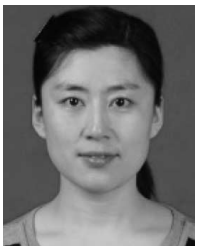

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